

Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics | Textbook: OpenStax Precalculus 2e (Jay Abramson, 2021)

COURSE AT A GLANCE

Review Content

REVIEW

1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties

UNIT LEARNING OBJECTIVES

- I can determine the domain and range of a function from a formula, graph, or table.
- I can describe and apply transformations of a parent function.
- I can analyze exponential functions using their equations, graphs, and contexts.
- I can analyze logarithmic functions using their equations, graphs, and contexts.
- I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

1.4 Composition of Functions; 1.7 Inverse Functions

UNIT LEARNING OBJECTIVES

- I can determine whether a function is even, odd, or neither.
- I can add, subtract, multiply, and divide functions and state the resulting domain.
- I can evaluate a composition from formulas, graphs, or tables.
- I can write a formula for the composition of two functions and determine its domain.
- I can decompose a function into a composition of simpler functions.
- I can determine whether a function has an inverse on a given domain.
- I can find and verify an inverse function algebraically.
- I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation; 3.8 Inverses and Radical Functions

UNIT LEARNING OBJECTIVES

- I can identify a power function and interpret the parameters in $y = kx^p$.
- I can construct a power function from two data points or a proportional relationship.
- I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
- I can recognize and model direct, inverse, and other power-variation relationships.
- I can solve equations involving integer or rational powers.
- I can build or select a power-function model for a context and interpret its parameters.
- I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

- I can predict the end behavior of a polynomial from its degree and leading coefficient.
- I can determine polynomial zeros and their multiplicities.
- I can construct a polynomial function from specified zeros or graphical features.
- I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
- I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.
- I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

3.7 Rational Functions; 12.1 Finding Limits: Numerical and Graphical Approaches; 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions; 3.9 Modeling Using Variation

UNIT LEARNING OBJECTIVES

- I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
- I can sketch and analyze a rational function using its algebraic and asymptotic features.
- I can interpret limit notation as a statement about function behavior near an input.
- I can estimate a limit from a graph or table.
- I can evaluate a limit algebraically using direct substitution or simplification.
- I can interpret an infinite limit as vertical-asymptotic behavior.
- I can use limit notation to describe behavior at infinity.
- I can analyze the long-run behavior of quotients involving polynomial, exponential, or logarithmic expressions.
- I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.
- I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

UNIT LEARNING OBJECTIVES

- I can evaluate and graph a sequence defined explicitly or recursively.
- I can write explicit and recursive formulas for arithmetic sequences.
- I can write explicit and recursive formulas for geometric sequences.
- I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
- I can create and interpret a sequence model for discrete change.
- I can interpret and use summation notation.
- I can calculate a finite arithmetic series.
- I can calculate a finite geometric series.
- I can determine whether an infinite geometric series converges and find its sum when it does.
- I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

12.1 Finding Limits: Numerical and Graphical Approaches

UNIT LEARNING OBJECTIVES

- I can determine one-sided limits and decide whether a two-sided limit exists.
- I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

UNIT LEARNING OBJECTIVES

- I can graph a plane curve defined by parametric equations with its orientation.
- I can eliminate a parameter to obtain a rectangular equation.
- I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates; 10.3 The Parabola

UNIT LEARNING OBJECTIVES

- I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
- I can use completing the square to rewrite a general-form conic equation in standard form.
- I can write, interpret, and graph implicit and parametric equations of a circle.
- I can write, interpret, and graph implicit and parametric equations of an ellipse.
- I can write, interpret, and graph implicit and parametric equations of a hyperbola.
- I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

**Conic Sections
(Continued)**

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.